Laeeq Aslam, PhD

Physics-guided Wind-speed Estimation | Wind-energy Forecasting | Mechanics-informed Modeling

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Professional Summary

Postdoctoral Research Fellow at Guangdong Technion–Israel Institute of Technology (GTIIT) with research focused on physics-guided wind-speed estimation from multirotor telemetry, atmospheric-flow evolution, and mechanics-informed data-driven modeling. Current work integrates drone telemetry, reference wind measurements, and interpretable machine learning to estimate ambient wind conditions from aerial platforms. Prior doctoral work focuses on short-term wind-speed prediction, spatio-temporal wind-field dynamics, integration of physical constraints, and embedded deployment for wind-energy applications. I combine machine learning with fluid mechanics and dynamical systems interpretation to develop robust forecasting methods for wind power engineering.

Publications & Research

Authored **12 journal articles (7 as first-author)** and **4 conference papers**, with **4** additional manuscripts **under review**.

- **Physics-Informed Ensemble Learning for City-Center Grid Cell Temperature Prediction During Thermal Extremes (Urban Climate, 2025 (IF: 6.4, JCR: Q1))**.
  - *Developed **PIEL-NET,** a **physics-informed hierarchical framework** for urban temperature forecasting that couples a simplified **advection-diffusion transport model** with **terrain-aware interaction** and spatio-temporal residual learning, thereby capturing wind-driven heat transport and mesoscale thermal evolution.*
  - *Designed a **Physics-Terrain Fusion Agent** that combines advective context from the city center and eight surrounding directions with a terrain-mediated coefficient model, then refines the forecast through residual ConvLSTM learning focused on hard extreme-regime errors.*
- **Physics-Informed Spatio-Temporal Network with Trainable Adaptive Feature Selection for Short-Term Wind Speed Prediction. (Computers & Electrical Engineering, 2025 (IF: 4.9, JCR: Q1)).**
  - *Developed **PISTNet**, a physics-informed spatio-temporal framework for short-term wind-speed prediction that couples adaptive feature selection with an **advection-equation-based physical model** to better represent wind-field transport and terrain-sensitive flow behavior.*
  - *Introduced selective physics-constrained training through **Adaptive Physics Penalty Loss**, improving forecasting accuracy across four sites and supporting **wind-power optimization and grid stability**.*
- **A Shallow Hybrid Model with Dynamic Bayesian Optimisation for Wind Speed Prediction on Memory-Constrained Devices. (Computers & Electrical Engineering (IF: 4.9, JCR: Q1)).**
  - *Developed a lightweight hybrid model for short-term wind-speed prediction that combines **FastICA**, adaptive LASSO, parallel **LSTM-TCN temporal modeling,** and RVFL-based refinement for resource-limited edge hardware.*
  - *Achieved a compact accuracy-efficiency balance for **edge wind-energy applications**, reducing parameter count by 38.04% while improving prediction accuracy and demonstrating deployment on Jetson Nano.*
- **Dynamic Optimization of Recurrent Networks for Wind Prediction on Edge Devices. (IEEE Access, 2025 (IF: 3.6, JCR: Q2)).**
  - *Developed a l**ocation-specific optimization framework** for LSTM, GRU, and TCN models under strict memory limits, targeting **decentralized and small-scale wind-power systems**.*
  - *Proposed **adaptive simulated annealing with memory-based rejection** to co-optimize forecast accuracy and model compactness, showing that optimal architecture selection depends on **local wind regimes and terrain-dependent variability**.*

- **Under Review**
  - *Physics-guided Wind-speed Estimation from Multirotor Telemetry using Centroid-gated Calibration and Residual Recurrent Correction (Under Review).*
  - *A Distribution Matching Framework for Zero-Shot Wind Speed Prediction on Edge Devices in Unseen Locations. (Under Review, Engineering Applications of Artificial Intelligence).*
  - *PiDR-Net: A Perception-Informed Deep Recurrent Network for Short-Term Wind Speed Predictions (Submitted, Applied Soft Computing).*
  - *Physics-Informed Extreme Wind Speed Prediction with a Multi-Gated Convolutional Recurrent Network VORTEX-Net (Under Review, Applied Energy).*

Research Experience

- **Postdoctoral Research Fellow – Guangdong Technion–Israel Institute of Technology (GTIIT), China** | March 2026 – Present
  - *Conducting research on **physics-guided wind-speed estimation** using multirotor **drone telemetry** and reference met-tower measurements.*
  - *Developing interpretable calibration and learning frameworks that relate drone attitude response to ambient wind conditions.*
  - *Leading model development, validation, and manuscript preparation for aerial wind sensing and scientific machine learning applications.*
- **PhD Researcher – Wind Power Forecasting and Mechanics-Informed Modeling** | Sept 2020 – Dec 2025
  - *Conducted doctoral research on **wind-power forecasting and atmospheric-flow behavior** for sustainable energy systems.*
  - *Developed **physics-informed spatio-temporal models** for short-term wind-speed prediction under multi-horizon forecasting settings.*
  - *Integrated **data-driven learning with physical and mechanical constraints** relevant to **wind-field evolution, flow consistency**, and engineering interpretability.*
  - *Established an experimental framework for **edge deployment** of wind prediction models on resource-constrained devices such as **NVIDIA Jetson Nano**.*
  - *Designed model architectures, **adaptive training losses, and feature-selection strategies** for robust forecasting across multiple global wind datasets.*
  - *Contributed to the **Key R&D Program of Hunan Province** (Project 2020WK2007) in support of wind-energy storage and grid-oriented optimization.*
- **AI/ML Consultant - Edge Computing & Computer Vision** | Bond and Built Pvt Ltd, Pakistan | July 2024 – March 2025
  - *Led the **applied research and development** for a real-time footwear analytics system, tackling challenges in large-scale edge deployment and multi-stream data processing.*
  - *Scaled a computer vision pipeline to process **50,000+ daily inferences across 60+ concurrent streams,** delivering actionable market intelligence and demonstrating robust system architecture.*
- **Machine Learning Engineer – Computer Vision & AI Deployment** | DLISION, Pakistan | May 2021 – Sept 2022
  - *Led **computer vision projects** focused on **semantic segmentation and real-time object detection**.*
  - *Optimized **UNet models with** convolutional and attention backbones, improving **segmentation accuracy by 3% against their baseline architectures** using **Focal Loss** parameters adjustment.*
- **Higher Education Research Assistant – AI for Healthcare** | International Islamic University, Pakistan| March 2019 – Feb 2020

- *Applied **Generative Adversarial Networks (GANs)** for **breast cancer detection**, improving classification accuracy.*
- *Conducted data augmentation techniques that **increased dataset diversity** and **enhanced model robustness**.*

- **Lecturer & Lab Engineer – Embedded Systems & AI** |iUSE School of Engineering and Management Sciences, Pakistan| Sept 2013 – Feb 2019
  - *Taking undergraduate courses related to **Embedded Systems**, **Programming,** and **Discrete Signal Processing**.*
  - *Designed and developed an **Instrumentation & Measurement Lab** for **sensor-based data acquisition in MATLAB**.*
  - *Led **student research projects** on **energy management, accident alerts, and secure electronic voting**.*

Education

- PhD in Control Science & Engineering | Central South University, China | Sept 2020 – Dec 2025.
- MS in Electronic Engineering| International Islamic University, Pakistan | Sept 2015 – Aug 2017.
- BS in Electronic Engineering| International Islamic University, Pakistan | Sept 2006 – Aug 2010.

Technical Skills

- **Machine Learning & AI:** Wind Energy Forecasting | Time-Series Modeling | Physics-Informed Learning
- **Programming & Tools:** Python | MATLAB | C++ | TensorFlow | PyTorch | OpenCV | Edge Impulse.
- **DevOps & Deployment:** Docker | AWS (EC2)| Git | Triton Inference Server.
- **Hardware & IoT:** Raspberry Pi | NVIDIA Jetson | Arduino Nano BLE Sense.
- **Data Analysis & Optimization:** Pandas | NumPy | Matplotlib | Seaborn | Feature Engineering

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Awards & Honors

- Gold Medalist – MS in Electronic Engineering | International Islamic University.
- Hunan Provincial Government Funding for Research & Development (**Project 2020WK2007**).
- Chinese Government CSC Scholarship for PhD Studies.

Professional Memberships & Certifications

- Registered Engineer, Pakistan Engineering Council (ELECTRO/22837)
- Member, Institute of Electrical and Electronics Engineers (IEEE)

References

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